A third cubic GHS cover screen: quadratic conjugate poles

Research audit | exact family census and an evaluated subgroup map

The construction below isolates the elliptic-quotient case whose two poles are conjugate over F_(q^3). It adds exact, provable counts beyond the triangle and rational-pole Hess screens. Earlier even-characteristic cubic classifications probably include this geometric case, so global family novelty remains unverified.

Construction

Let q=2^m, k=F_q, K=F_(q^3), k2=F_(q^2), and L=F_(q^6). Choose t in k2 with t+t^q=1 and nonzero A in W=ker Tr_(L/k2). Set s=A A^(q^3), b=s^2, and f_i(x)=A^(q^i)/(x+t^(q^i))+A^(q^(i+3))/(x+t^(q^(i+3))) for i=0,1,2. The f_i lie in K(x), cycle under Frobenius, and sum to zero. Their V4 compositum descends to a genus-three curve over k. Its first elliptic quotient is the trace-one twist of Y^2+XY=X^3+aX^2+b.

Exact all-q bound

```
|Norm(W*)| = (q^3+q-2)/2
N_new(q) = (q^3-q^2-4)/4                m odd
         = (q^3-q^2+4q-4)/4            m even
Union(q) = 3q^3/2-2                   m odd
         = 3q^3/2+q-2                 m even
```

N_new counts models in this quadratic-pole screen outside the triangle screen. Union counts the triangle, Hess rational-pole, and quadratic-pole screens among 2(q^3-1) normalized ordinary models. These are exact cover-screen counts, not a global count of vulnerable curves or an attack-cost bound.

Proof in one paragraph

W is the k2-extension of the two-dimensional k-space ker Tr_(K/k). Multiplication Sym^2(W) to L is an isomorphism, so projective norm values are Frobenius orbits on P^1(k2). There are (q^2+q+2)/2 such orbits. Exactly two have zero relative trace when m is even and none when m is odd. Scalar absolute-trace counts then give N_new. The full argument and independently enumerated field values are in REPORT.md and population.py.

Exact q=16 source and measured evidence

Field L=F2[z]/(z^24+z^16+z^15+z^14+z^13+z^10+z^9+z^7+z^5+z^3+1). Decimal coefficients below are binary polynomial bitmasks in the basis 1,z,...,z^23. K is its unique 2^12 subfield.

```
E/K: Y^2+XY = X^3 + 5522909 X^2 + 9159130
#E(K) = 4054 = 2*2027
G=(9159131,11791722)
Q=(744530,10466263)=7G
t=16191718, A=2832837
```

The canonical curve over F16 is a smooth nonhyperelliptic quartic with L-polynomial 1-43T^3+4096T^6. Its exact equation and quotient rational map are in canonical_4054.json and quotient_map_4054.json. The evaluated traced correspondence satisfies Phi(G) nonzero, [2027]Phi(G)=0, Phi(Q) nonzero, [2027]Phi(Q)=0, and Phi(Q)=[7]Phi(G).

Population and costs

q	Norm values	Added models	Three-screen union / all
2	4	0	10 / 14
4	33	15	98 / 126
8	259	111	766 / 1022
16	2055	975	6158 / 8190

The full F256 Magma run verified the divisor orders, found 260 relations over 201 factor-base points, solved the matrix, and returned 7 mod 2027. Its proof-heavy job took 67.239 s. A separate timed run omitted those already checked assertions: setup 0.190 CPU s, index recovery 0.170 CPU s, and complete server job 0.459 s.

On the same source subgroup and target, explicit Magma Floyd rho solved 100/100 runs, median 168 group step calls, in 0.160 CPU s total (0.00160 s per solve on average). The measured Magma GHS core is about 225 times that average rho solve. Local Sage source-map setup and evaluation add 15.15 s and 5.04 s on another machine, so they are reported separately. This q=16 example shows no GHS-over-rho advantage.

Prior-work boundary

The order-4054 Jacobian isogeny class was already present in the repository and in LMFDB as 3.16.a_a_abr. Shimura, Momose, and Chao (SCIS 2010) studied cubic even-characteristic weak covers; a 2015 survey already attributes a three-quarters density to that work. Murai et al. (SCIS 2022) announced a corrected classification under the isogeny condition. The exact source equation and map are new to this audit; global family and method novelty remain unverified.

References and reproducibility

Primary sources: https://www.iwsec.org/scis/2022/_abst/1A3-2.pdf ; https://arxiv.org/abs/0905.0546 ; https://arxiv.org/abs/2012.07173 . Scripts, Magma receipts, the checked Lean arithmetic, and the full proof are in this research directory.